Passenger Intercom System (S.O.O. 6466) – General Data

1. General

The Passenger Intercom System, (if installed) is part of the audio integration system, refer to 23-50-00, Audio Integration – General Data. The Passenger Intercom System is only available to an 8-seat passenger configuration. It allows passengers to communicate among themselves and with the flight compartment crew.

2. Description

A. Passenger Intercom System

The Passenger Intercom System consists of two audio controllers, eight headset jacks and a VOX control knob.

1) Audio Controllers (Jupiter JA95-R04)

There are two dedicated Jupiter JA95-R04 Audio Controllers used to control Lemo headset jacks. JA95-R04 #1 controls Group 1 headsets installed on the RH side of the passenger cabin. JA95-R04 #3 controls Group 3 headsets installed on the LH side of the passenger cabin.

28 Volt DC power to both Audio Controllers is protected by a 3 amp PAX ICS (6Q) (Right Bus) circuit breaker located on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

2) Lemo Headset Jacks

There are two rows of four (4) Lemo headset jacks installed along the right-hand and left-hand sidewall panel forward of each passenger seat. These jacks are only compatible with A20 Bose headsets.

28 Volt DC power to all Lemo headsets is protected by a 2 amp PAX HDST (6H) (Right Bus) circuit breaker located on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

3) VOX Control Knob

A VOX control knob is installed on the cockpit bulkhead behind the pilot's seat (on the map light enclosure).

B. Operation

The Passenger Intercom System is interfaced with the existing co-pilot panel for flight crew communication with the passengers.

When the audio control panel ICS "ALL" is selected, both Jupiter JA95-R04 audio controllers act as ICS tie for all the passengers and the flight compartment crew.

The audio controllers are configured for headset extension (Type B) where the mic for each passenger headset is live and does not require PTT (press-to-talk) switch. The passengers can communicate to each other regardless of the cockpit selection on the audio control panel unless the VOX control knob is on the PTT position (fully clockwise).

The Passenger Intercom System interfaces to the following aircraft systems:

1) Audio Control Panel System (refer to 23-53-00, Audio Control Panel – General Data).

3. Adjustment/Test

A. Audio Controller Set-up/Adjustment

1) General

Note These procedures are for the set-up/adjustment of the audio controller box (Jupiter JA95-R04), when replaced with new unit.

a) Ensure that PAX ICS (6Q) and PAX HDST (6H) circuit breaker is closed.

b) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

c) Wait several minutes for the aircraft electrical systems to initialize.

d) Connect the maintenance laptop computer to the Jupiter JA95-R04 Audio Controller using cables JA99-001 and CAB-USB-0002 from an available computer COM port. Start the Audio Controller software ProCS™ V0.16.6 or higher. (Refer to Pro CS Product Configuration Software Installation and Operating Manual for detailed procedures).

e) Select appropriate computer COM port.

Note If communication between the audio controller and the laptop is not established, the ProCS™ remains on demo mode.

f) Check configuration settings as follows:

1 Select Headset Expansion Type B. Configure connector pins as required, refer to PSM 1-64-2W, 44-12-00, 8 Passenger Headset Jacks Installation (S.O.O. 6466).

2 Check off “Mute RX Audio" and "Mute ICS Audio" for audio muting.

3 Do ICS and Headset Expansion Settings.

4 Select VOX control "External".

g) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Function Test Passenger Intercom System

1) General

These procedures are for the function test of the Passenger Intercom System.

2) Function Test

a) Ensure that PAX ICS (6Q) and PAX HDST (6H) circuit breaker is closed.

b) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

c) Wait several minutes for the aircraft electrical systems to initialize.

d) On the audio control panel, press ICS rocker switch to select ALL mode.

e) Begin with Group 1, Headset #1 plugged in. Confirm that there is a clear communication between the flight compartment crew and the passenger. Verify that there are no unusual hums, buzzes, or other background noise.

f) Verify proper volume control operation. Turn the VOX control knob to maximum clockwise position (PTT) and verify that passenger’s microphones are off. Turn the VOX control knob back to the desired volume.

g) On the pilot’s audio control panel, press ICS rocker switch to select CREW mode.

h) Plug in the Group 2 Headset #1. Check for correct intercom operation and communication between 2 passengers.

i) Verify that passengers cannot communicate with the flight compartment crew.

j) Verify that passengers can communicate with each other without interrupting the flight crew.

k) Plug in the remaining headset locations. Verify each location for proper intercom operation. Verify volume control operation applies to all passenger seat locations. Verify that there are no unusual hums, buzzes or other background noise during the entire operation.

l) Verify that passengers can communicate with the flight compartment crew by selecting ICS - "ALL".

m) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

4. Removal/Installation

WARNING **INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECTS OF LIGHTNING (IEL). ENSURE INTEGRITY OF THE HIRF/IEL PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:**

CHASSIS GROUNDING – ALL RACK MOUNT/REMOTE MOUNT UNITS SHALL BE ELECTRICALLY BONDED TO THE AIRFRAME. ENSURE THAT THE MATING SURFACES BETWEEN THE LRU MOUNTING TRAY (OR LRU MOUNTING FEET IF A TRAY IS NOT USED) PROVIDE A LOW IMPEDANCE ELECTRICAL PATH. ALL MATING SURFACES MUST BE FREE OF ALL PAINT AND OTHER NONCONDUCTIVE ELEMENTS. REFER TO BONDING REQUIREMENTS IN 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES – GENERAL DATA.

SHIELD GROUNDS – VERIFY SHIELD GROUNDING AS PER REQUIREMENTS IN PSM 1-64-2W.

A. Removal and Installation of the Audio Controller (Jupiter JA95-R04)

1) General

These procedures are for the removal and installation of the Audio Controller.

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Open PAX ICS (6Q) circuit breaker on the avionics circuit breaker panel and fit with a safety clip.

c) Gain access to the cabin roof right hand side at STA 125.55 and cabin sidewall left hand side at STA 172.00.

d) Disconnect the electrical connectors on the back of the unit.

e) Remove the (4) screws securing the unit to the bracket they are installed on.

f) Remove unit from the aircraft.

Note Stow connectors as required.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the PAX ICS (6Q) circuit breaker on the avionics circuit breaker panel is opened and fitted with a safety clip.

c) Align the unit to the mounting holes on the bracket where they will be installed.

d) Secure the unit with (4) mounting screws.

e) Reconnect connectors to the unit. Ensure connections are secure.

f) Remove the safety clips and close PAX ICS (6Q) circuit breaker on the avionics circuit breaker panel.

g) If new unit is installed, complete Audio Controller Set-up/Adjustment as per Para 3.A.

h) Test the Passenger Intercom System as per Para 3.B.

5. Troubleshooting

A. Passenger Intercom System

1) If the Passenger Intercom System is deemed inoperative, check the following:

a) Ensure PAX ICS (6Q) and PAX HDST (6H) circuit breakers are closed.

b) Visually inspect the audio controller unit for damaged wires or contacts.

c) Visually inspect the Lemo headset jacks for damaged wires or contacts.

d) Ensure that the audio controllers and headset jack disconnect connectors are properly mated and secure.

e) If troubleshooting at the wire level is required, refer to the applicable wiring diagrams in Wiring Diagrams Manual PSM 1-64-2W.

f) If repair or service is required, send to appropriately rated avionics repair facility.